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United States Patent	12386107
Kind Code	B2
Date of Patent	August 12, 2025
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Optical system

Abstract

A prism includes a first intermediate imaging position at which a component in a first direction of the light flux is imaged and a second intermediate imaging position at which a component in a second direction orthogonal to the first direction of the light flux is imaged, different from the first intermediate imaging position. At least one of the first intermediate imaging position and the second intermediate imaging position lying within a first range from the second surface between the first surface and the second surface or lying within a second range from the second surface between the second surface and the third surface. The first range has a length less than one-half of an optical path length from the first surface to the second surface, and the second range has a length less than one-half of the optical path length from the second surface to the third surface.

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Appl. No.: 17/970942

Filed: October 21, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230047387 A1	Feb. 16, 2023

Foreign Application Priority Data

JP	2020-077618	Apr. 24, 2020
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Related U.S. Application Data

continuation parent-doc WO PCT/JP2020/047506 20201218 PENDING child-doc US 17970942

Publication Classification

Int. Cl.: G02B5/04 (20060101)

U.S. Cl.:

CPC G02B5/04 (20130101);

Field of Classification Search

CPC: G02B (5/04); G02B (5/10); G02B (17/0804); G02B (17/0812); G02B (17/082); G02B (17/0828); G02B (17/0836); G02B (17/0856); G02B (17/086)

USPC: 359/365; 359/366; 359/727; 359/730; 359/833; 359/834; 359/837; 359/858; 359/861

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2769373	12/1955	Bouwers	N/A	N/A
4247772	12/1980	Lavelle et al.	N/A	N/A
4948233	12/1989	Maruyama	N/A	N/A
6134051	12/1999	Hayakawa	359/640	G02B 17/086
2002/0101666	12/2001	Togino	359/720	G02B 15/04
2002/0163742	12/2001	Togino	N/A	N/A
2015/0061975	12/2014	Komatsu et al.	N/A	N/A
2016/0334637	12/2015	Saisho et al.	N/A	N/A
2018/0259777	12/2017	Uemura	N/A	G02B 27/0172
2021/0364784	12/2020	Kuzuhara	N/A	G02B 17/0816
2021/0373304	12/2020	Kuzuhara	N/A	G02B 26/10
2021/0373305	12/2020	Kuzuhara	N/A	G02B 17/04

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
354843	12/1930	GB	N/A
2000-231060	12/1999	JP	N/A
2015-148664	12/2014	JP	N/A
2018/166921	12/2017	WO	N/A
2019/188070	12/2018	WO	N/A

OTHER PUBLICATIONS

International Preliminary Report on Patentability issued Nov. 3, 2022 in International Application No. PCT/JP2020/047506. cited by applicant

Extended European Search Report issued Oct. 13, 2023 in corresponding European Patent Application No. 20931807.0. cited by applicant

International Search Report issued on Feb. 9, 2021 in International Application No. PCT/JP2020/047506, with English translation. cited by applicant
Office Action (Notice of Reasons for Refusal) issued May 7, 2024 in Japanese Application No. 2022-516843, with English translation. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation application of International Application No. PCT/JP2020/047506, with an international filing date of Dec. 18, 2020, which claims priority of Japanese Patent Application No. 2020-77618 filed on Apr. 24, 2020, the content of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Technical Field

(1) The present disclosure relates to an optical system using a prism.

2. Description of Related Art

(2) In an optical system, reflecting light with a prism is better for miniaturization of the optical system than reflecting it with a mirror. In the case of using a prism, an intermediate image of light may be formed within the interior of the prism. For example, also in a prism of JP 2000-231060 A, there is formed an intermediate image of light travelling inside the prism.

SUMMARY

(3) In cases where scratch or dust within the prism is present at an intermediate imaging point of light entering the prism, part of light may possibly disappear. A reflection surface of the prism has greater influence on light due to manufacturing error than a refraction surface does.

(4) The present disclosure provides an optical system that comprises a prism reducing the influence of scratch inside the prism as well as reducing the influence of manufacturing error.

(5) An optical system of the present disclosure comprises a prism that includes a first surface, a second surface, and a third surface, with a light flux from the first surface being reflected by the second surface to head for the third surface, the prism having a first intermediate imaging position at which a component in a first direction of the light flux incident inside is imaged and a second intermediate imaging position at which a component in a second direction orthogonal to the first direction of the light flux is imaged, different from the first intermediate imaging position, at least one of the first intermediate imaging position and the second intermediate imaging position lying within a first range from the second surface between the first surface and the second surface or lying within a second range from the second surface between the second surface and the third surface, the first range having a length less than one-half of an optical path length from the first surface to the second surface, the second range having a length less than one-half of the optical path length from the second surface to the third surface.

(6) An optical system of the present disclosure comprises a prism that includes a first surface, a second surface, a third surface, and a fourth surface, with a light flux from the first surface being reflected by the second surface to head for the third surface, the light flux from the second surface being reflected by the third surface to head for the fourth surface, the prism having a first intermediate imaging position at which a component in a first direction of the light flux incident inside is imaged, and a second intermediate imaging position at which a component in a second direction orthogonal to the first direction of the light flux is imaged, different from the first

intermediate imaging position, one of the first intermediate imaging position and the second intermediate imaging position lying within a first range from the second surface between the first surface and the second surface or lying within a second range from the second surface between the second surface and the third surface, the other of the first intermediate imaging position and the second intermediate imaging position lying within the second range from the third surface between the second surface and the third surface or lying within a third range from the third surface between the third surface and the fourth surface, the first range having a length less than one-half of an optical path length from the first surface to the second surface, the second range having a length less than one-half of the optical path length from the second surface to the third surface, the third range having a length less than one-half of the optical path length from the third surface to the fourth surface.

(7) The optical system in the present disclosure enables provision of an optical system that comprises a prism reducing the influence of scratch inside the prism as well as reducing the influence of manufacturing error.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a sectional view showing a configuration of an optical system in a first embodiment.
- (2) FIG. 2 is a diagram showing a pupil diameter of laser light immediately after irradiation from a laser element in the first embodiment.
- (3) FIG. 3 is a diagram showing respective intermediate imaging positions of an X-component and a Y-component of laser light.
- (4) FIG. 4 is a diagram showing a pupil diameter of laser light at a first intermediate imaging position (Px).
- (5) FIG. 5 is a diagram showing a pupil diameter of laser light at a second intermediate imaging position (Py).
- (6) FIG. 6 is a diagram showing a pupil diameter of laser light exiting a prism.
- (7) FIG. 7 is a sectional view showing a configuration of an optical system in a variant of the first embodiment.
- (8) FIG. 8 is a sectional view showing a configuration of an optical system in a variant of the first embodiment.
- (9) FIG. 9 is a sectional view showing a configuration of an optical system in a variant of the first embodiment.
- (10) FIG. 10 is a sectional view showing a configuration of an optical system in a variant of the first embodiment.
- (11) FIG. 11 is a sectional view showing a configuration of an optical system in a second embodiment.
- (12) FIG. 12 is a diagram showing respective intermediate imaging positions of the X-component and Y-component of laser light.
- (13) FIG. 13 is a sectional view showing a configuration of an optical system in a variant of the second embodiment.
- (14) FIG. 14 is a diagram showing respective intermediate imaging positions of the X-component and Y-component of laser light.
- (15) FIG. 15 is a sectional view showing a configuration of an optical system in a variant of the second embodiment.
- (16) FIG. 16 is a diagram showing respective intermediate imaging positions of the X-component and Y-component of laser light.
- (17) FIG. 17 is a sectional view showing a configuration of an optical system in a third

embodiment.

(18) FIG. **18** is a diagram showing respective intermediate imaging positions of the X-component and Y-component of laser light.

(19) FIG. **19** is a sectional view showing a configuration of an optical system in a variant of the second embodiment.

DETAILED DESCRIPTION

(20) Embodiments will hereinafter be described in detail with proper reference to the drawings. In some cases, however, more detailed description than necessary may be omitted. For example, detailed description of already well-known matters or duplicate description for substantially the same configuration may be omitted. This is to avoid the following description from becoming unnecessarily redundant, to facilitate the understanding of those skilled in the art.

(21) It is to be noted that the inventor(s) provides the accompanying drawings and the following description for those skilled in the art to fully understand the present disclosure, and that it is not intended to limit thereby the subject matter described in the claims.

First Embodiment

(22) Referring to FIGS. **1** to **6**, a first embodiment will be described below. In this embodiment, as shown in FIG. **2**, for example, X-direction is a major axis direction of a pupil diameter of laser light R emitted from a laser element **3**, and Y-direction is a minor axis direction of the pupil diameter of laser light R emitted from the laser element **3**. X-direction and Y-direction are orthogonal to each other. An XYZ coordinate system is indicated as coordinates of an optical system. X-direction, Y-direction, and Z-direction are orthogonal to one another.

(23) [1-1. Configuration]

(24) FIG. **1** is a sectional view showing a configuration of an optical system **1** according to the present disclosure. The optical system **1** includes a laser element **3** as a light source, and a prism **5**.

(25) The laser element **3** is, for example, a semiconductor laser. Laser light R irradiated from the laser element **3** is parallel light with different pupil diameters in X-direction as a first direction and in Y-direction as a second direction. For example, as shown in FIG. **2**, laser light R immediately after irradiation from the laser element **3** has a pupil diameter **3a** of an oval shape extending in X-direction. Laser light R irradiated from the laser element **3** is incident on an incident surface **11** of the prism **5**. Laser light R has a plurality of wavelengths (wavelength ranges) so as to have colors of, for example, R(red), G(green), and B(blue). The laser element **3** may emit laser light R as one light flux in which R, G, and B lights are mixed, or may emit laser light R in sequence for each color light flux.

(26) The prism **5** has an incident surface **11** as a first surface, a first reflection surface **13** as a second surface, and an exit surface **15** as a third surface. The first reflection surface **13** is arranged at a position on an optical path from the incident surface **11** to the exit surface **15**.

(27) Laser light R emitted from the laser element **3** is incident through the incident surface **11** into the prism **5**. The incident surface **11** and the first reflection surface **13** face each other so that laser light R incident from the incident surface **11** reduces the pupil diameter **3a** in Y-direction and is imaged at a second intermediate imaging position P_y lying on the optical path between the incident surface **11** and the first reflection surface **13**, after which it again increases the pupil diameter **3a** in Y-direction, to be reflected by the first reflection surface **13** into the interior of the prism **5**.

(28) The incident surface **11** has, for example, an outward protruding convex shape. The incident surface **11** may be a rotation asymmetric surface. For example, it may be of a free curved surface shape with different curvatures in X-direction and Y-direction. By rendering the shape of the incident surface **11** convex, laser light R incident from the incident surface **11** can be refracted inward. Since laser light R can travel in the prism **5** while suppressing its spread in this manner, the prism **5** can be reduced in size. The incident surface **11** may have a refractive power in Y-direction smaller than that in X-direction. This enables suppression of chromatic aberration in Y-direction generated on the incident surface **11**.

(29) The prism **5** is made of, for example, resin or glass. The prism **5** has a refractive index n satisfying a relationship of $n < 1.8$. This enables the influence of manufacturing error of a refraction surface of the prism **5** to be reduced. The refraction surface of the prism **5** is, for example, the incident surface **11**. The prism **5** has an Abbe number vd satisfying a relationship of $vd > 40$. This achieves reduction in chromatic aberration of light incident on the prism **5**.

(30) Laser light **R** incident from the incident surface **11** travels reducing the pupil diameter $3a$ in X-direction, and laser light **R** reflected by the first reflection surface **13** is intermediately imaged at a first intermediate imaging position P_x lying on the optical path between the first reflection surface **13** and the exit surface **15**, after which it again increases the pupil diameter $3a$ in X-direction, to be emitted from the exit surface **15** to the outside of the prism **5**.

(31) The first reflection surface **13** has different curvatures in X-direction and Y-direction. The first reflection surface **13** therefore has a free curved surface shape. The first reflection surface **13** may be eccentric with respect to the incident light. This enables the optical path of the incident light to be split without using an optical element such as a beam splitter.

(32) The exit surface **15** of the prism **5** may also have a similar configuration to that of the incident surface **11**. The exit surface **15** may have an outward protruding convex shape. This enables the light flux traveling in the prism **5** while suppressing the spread of laser light **R** to be focused outside the prism **5**. The curvatures in X-direction of the incident surface **11** and the exit surface **15** may have symmetry. The exit surface **15** may have a refractive power in X-direction smaller than that in Y-direction. The exit surface **15** may be of a free curved surface shape with different curvatures in X-direction and Y-direction.

(33) In the optical system **1** of this embodiment, the incident surface **11** of the prism **5**, the first reflection surface **13** of the prism **5**, and the exit surface **15** of the prism **5** are arranged in the mentioned order on the optical path from the laser element **3**.

(34) As shown in FIG. **3**, the optical system **1** has the first intermediate imaging position P_x in X-direction of the light flux of laser light **R** between the first reflection surface **13** of the prism **5** and the exit surface **15** of the prism **5**. That is, laser light **R** is intermediately imaged by the incident surface **11** or the first reflection surface **13**.

(35) Since R_x as an X-direction component of laser light **R** and R_y as a Y-direction component have different respective focal lengths, the first intermediate imaging position P_x of the X-component R_x of laser light **R** and the second intermediate imaging position P_y of the Y-component R_y can be arranged at different positions. Due to the X-component R_x and the Y-component R_y having different respective focal lengths, respective magnifications are also different when emitted from the exit surface **15** of the prism **5**. That is, the optical system **1** has optical magnifications different in X-direction and Y-direction. For example, since in this embodiment the focal length is greater in Y-direction than in X-direction, the optical magnification is greater in Y-direction than in X-direction.

(36) For example, let M_x be the horizontal magnification in X-direction and let M_y be the horizontal magnification in Y-direction,

(37) a relationship of

(38) $M_y/M_x > 1.1$

(39) is satisfied. This results in an increased optical path length difference between the first intermediate imaging position P_x and the second intermediate imaging position P_y , allowing alleviation of the influence of scratch or dust inside the prism **5**.

(40) By satisfying a relationship of $M_y/M_x > 1.3$, either the first intermediate imaging position P_x or the second intermediate imaging position P_y can be arranged on the optical path between the incident surface **11** and the first reflection surface **13**, while either the first intermediate imaging position P_x or the second intermediate imaging position P_y can be arranged on the optical path between the first reflection surface **13** and the exit surface **15**, allowing further alleviation of the influence of scratch or dust inside the prism **5**.

(41) By satisfying a relationship of $M_y/M_x > 1.5$, the optical path length difference between the first intermediate imaging position P_x and the second intermediate imaging position P_y further increases, achieving further improvement in the degree of freedom of arrangement of the first intermediate imaging position P_x and the second intermediate imaging position P_y .

(42) The first intermediate imaging position P_x of the light flux of laser light R in X-direction does not intersect, at the same position, the light flux of laser light R in Y-direction orthogonal to X-direction. That is, the first intermediate imaging position P_x of the X-component R_x of laser light R does not lie on the same position as that of the second intermediate imaging position P_y of the Y-component R_y of laser light R. Consequently, as shown in FIG. 4, a pupil diameter $3b$ of laser light R in an intermediate image formed at the first intermediate imaging position P_x has a rectilinear shape extending in Y-direction. As a result, even if there is dust or scratch at the first intermediate imaging position P_x , since the Y-component R_y of laser light R is not imaged, the pupil diameter $3b$ of laser light R can be prevented from disappearing.

(43) As shown in FIG. 3, within the prism 5, the second intermediate imaging position P_y of the Y-component R_y of laser light R lies, on the optical path, closer to the incident surface 11 than the first intermediate imaging position P_x does where the X-component R_x of laser light R is imaged. As shown in FIG. 5, a pupil diameter $3c$ of laser light R in an intermediate image formed at the second intermediate imaging position P_y also has a rectilinear shape extending in X-direction. As a result, even if there is dust or scratch at the second intermediate imaging position P_y , since the X-component R_x of laser light R is not imaged, the pupil diameter $3c$ of laser light R can be prevented from disappearing. Due to the optical system 1 having a greater optical magnification in Y-direction than in X-direction, a pupil diameter $11d$ of laser light R emitted from the exit surface 15 is formed in a circular shape, as shown in FIG. 6. The first intermediate imaging position P_x may lie, on the optical path, closer to the incident surface 11 than the second intermediate imaging position P_y does.

(44) A relationship of

$$(45) 0.1 < (\phi_x \times 1 \times \phi_y 1) / (\phi_x 2 \times \phi_y 2) < 0.8$$

(46) is established between: a first emission pupil diameter $\phi_x 1$ in X-direction and a second emission pupil diameter $\phi_y 1$ in Y-direction of the light flux emitted from the laser element 3; and a first projection pupil diameter $\phi_x 2$ in X-direction and a second projection pupil diameter $\phi_y 2$ in Y-direction of the light flux passing through the exit surface 15 of the prism 5. By satisfying this relationship, the spot size at the intermediate imaging positions P_x and P_y increases, enabling effective alleviation of the influence of dust or scratch within the interior of the prism 5.

(47) As shown in FIGS. 1 and 3, the second intermediate imaging position P_y lies, toward the first reflection surface 13, within a first range $Rg1$ between the incident surface 11 and the first reflection surface 13. The first range $Rg1$ has a length less than one-half of an optical path length $L1$ from the incident surface 11 to the first reflection surface 13. The first range $Rg1$ may be of a length less than one-quarter or a length less than one-eighth of the optical path length $L1$ from the incident surface 11 to the first reflection surface 13. The closer the distance between the second intermediate imaging position P_y and the first reflection surface 13 is, the smaller the size (footprint size) of the light flux at the first reflection surface 13 can be. The smaller the size of the light flux at the first reflection surface 13 is, the less the influence of manufacturing error of the first reflection surface 13 on the light flux can be. Accordingly, if the first range $Rg1$ has a length less than one-quarter, instead of less than one-half, of the optical path length $L1$ from the incident surface 11 to the first reflection surface 13, the influence of manufacturing error can be further reduced, and if it has a length less than one-eighth instead of less than one-quarter, the influence of manufacturing error can be even further reduced. An example of the influencing manufacturing error is, for example, a curvature error of the first reflection surface 13.

(48) The first intermediate imaging position P_x lies within a second range $Rg2$ from the first reflection surface 13 between the first reflection surface 13 and the exit surface 15. The second

range Rg2 has a length less than one-half of an optical path length L2 from the first reflection surface 13 to the exit surface 15. The second range Rg2 may be of a length less than one-quarter or a length less than one-eighth of the optical path length L2 from the first reflection surface 13 to the exit surface 15. The closer the distance between the first intermediate imaging position Px and the first reflection surface 13 is, the smaller the size (footprint size) of the light flux at the first reflection surface 13 can be. The smaller the size of the light flux at the first reflection surface 13 is, the less the influence of manufacturing error of the first reflection surface 13 on the light flux can be. Accordingly, if the second range Rg2 has a length less than one-quarter, instead of less than one-half, of the optical path length L2 from the first reflection surface 13 to the exit surface 15, the influence of manufacturing error can be more reduced, and if it has a length less than one-eighth instead of less than one-quarter, the influence of manufacturing error can be even further reduced.

(49) As shown in FIGS. 1 and 3, in the case that the first reflection surface 13 lies on the optical path between the first intermediate imaging position Px and the second intermediate imaging position Py, the first intermediate imaging position Px and the second intermediate imaging position Py can each be arranged closer to the first reflection surface 13. Although the reflection surface is a member having a high error sensitively, the influence of manufacturing error of the first reflection surface 13 can be more reduced by arranging the first intermediate imaging position Px and the second intermediate imaging position Py closer to the vicinity of the first reflection surface 13. The influence of dust or scratch of the prism 5 can be more reduced by arranging the first intermediate imaging position Px and the second intermediate imaging position Py within their respective ranges different from the first reflection surface 13.

(50) Let ΔP be the optical path length difference between the optical path length of the first intermediate imaging position Px from the incident surface 11 and the optical path length of the second intermediate imaging position Py from the incident surface 11,

(51) a relationship of

(52) $\Delta P/L1 > 0.05$

(53) is satisfied. It is consequently possible to properly set the optical path length difference between the first intermediate imaging position Px and the second intermediate imaging position Py to alleviate the influence of scratch or dust inside the prism 5.

(54) Although in this embodiment both the first intermediate imaging position Px and the second intermediate imaging position Py lie in the vicinity of the first reflection surface 13, at least one of the first intermediate imaging position Px and the second intermediate imaging position Py may lie in the vicinity of the first reflection surface 13. It is thus possible to decrease the size of the light flux reflected by the first reflection surface 13 to thereby reduce the influence of manufacturing error of the first reflection surface 13 on the light flux reflected by the first reflection surface 13.

(55) Although in this embodiment the first intermediate imaging position Px lies between the first reflection surface 13 and the exit surface 15 while the second intermediate imaging position Py lies between the incident surface 11 and the first reflection surface 13, reverse arrangement is also possible. That is, the first intermediate imaging position Px may lie between the incident surface 11 and the first reflection surface 13, while the second intermediate imaging position Py may lie between the first reflection surface 13 and the exit surface 15.

(56) Although in this embodiment the first intermediate imaging position Px lies between the first reflection surface 13 and the exit surface 15 while the second intermediate imaging position Py lies between the incident surface 11 and the first reflection surface 13, this is not limitative. As in an optical system 1A shown in FIG. 7, both the first intermediate imaging position Px and the second intermediate imaging position Py may lie, between the incident surface 11 and the first reflection surface 13 of a prism 5A, within the first range Rg1 from the first reflection surface 13. As in an optical system 1B shown in FIG. 8, both the first intermediate imaging position Px and the second intermediate imaging position Py may lie, between the incident surface 11 and the first reflection surface 13 of a prism 5B, within the second range Rg2 from the first reflection surface 13.

(57) As in an optical system **10** shown in FIG. **9**, a lens element **7** having different condensing actions in X-direction and Y-direction of the light flux may be arranged between the laser element **3** and the incident surface **11** of a prism **5C**. The lens element **7** is, for example, a cylindrical lens having different curvatures for X-direction and Y-direction of the light flux. The light flux exiting the lens element **7** is incident on the incident surface **11**. Since the optical system **1** includes the lens element **7**, the first intermediate imaging position **Px** and the second intermediate imaging position **Py** can be arranged at proper positions, achieving greater flexibility in prism design. By processing the lens element **7** rather than by processing the shape of the incident surface **11**, the accuracy can be improved, facilitating the design of the intermediate imaging positions within the prism **5**. The prisms **5A**, **5B**, and **5C** have the same configuration as that of the prism **5** except for the respective points described above.

(58) [1-2. Effects, etc.]

(59) The optical system **1** according to the first embodiment is an optical system that includes the prism **5** having the incident surface **11** as the first surface, the first reflection surface **13** as the second surface, and the exit surface **15** as the third surface, with the light flux from the incident surface **11** being reflected by the first reflection surface **13** to head for the exit surface **15**. The prism **5** has: the first intermediate imaging position **Px** at which the component in X-direction as the first direction of the light flux incident inside is imaged; and the second intermediate imaging position **Py** at which the component in Y-direction as the second direction orthogonal to the first direction of the light flux is imaged, different from the first intermediate imaging position **Px**. At least one of the first intermediate imaging position **Px** and the second intermediate imaging position **Py** lies within the first range **Rg1** from the first reflection surface **13** between the incident surface **11** and the first reflection surface **13**, or lies within the second range **Rg2** from the first reflection surface **13** between the first reflection surface **13** and the exit surface **15**. The first range **Rg1** has the length less than one-half of the optical path length **L1** from the incident surface **11** to the first reflection surface **13**. The second range **Rg2** has the length less than one-half of the optical path length **L2** from the first reflection surface **13** to the exit surface **15**.

(60) Since the two intermediate imaging positions are formed within the prism **5** based on such a configuration, partial loss of the light flux due to scratch or dust inside the prism **5** can be reduced. Since at least one of the first intermediate imaging position **Px** and the second intermediate imaging position **Py** lies in the vicinity of the first reflection surface **13**, it is possible to diminish the size (footprint size) of the light flux at the first reflection surface **13** to reduce the influence of manufacturing error of the first reflection surface **13** on the light flux.

(61) In the case that both the first intermediate imaging position **Px** and the second intermediate imaging position **Py** lie in the vicinity of the first reflection surface **13**, the size (footprint size) of the light flux at the first reflection surface **13** can be even smaller, enabling further reduction in the influence of manufacturing error of the first reflection surface **13** on the light flux.

(62) Although in this embodiment the prism **5** has only the first reflection surface **13** as the reflection surface, it may have one or more other reflection surfaces on the optical path between the first reflection surface **13** and the exit surface **15**. In this manner, the prism **5** may have at least two or more reflection surfaces.

(63) As in an optical system **1D** shown in FIG. **10**, if the prism **5** has a second reflection surface **17** on the optical path between the first reflection surface **13** and the exit surface **15**, then the first reflection surface **13**, the second reflection surface **17**, and the exit surface **15** may act as the first surface, the second surface, and the third surface, respectively. A prism **5D** has the same configuration as that of the prism **5D** except for this point. If the prism **5** includes three or more reflection surfaces, then the first reflection surface **13**, another reflection surface, and yet another reflection surface may act as the first surface, the second surface, and the third surfaces, respectively.

Second Embodiment

(64) Referring next to FIGS. **11** and **12**, a second embodiment will be described. FIG. **11** is a diagram showing a configuration of an optical system **1E** in the second embodiment. FIG. **12** is a diagram showing respective intermediate imaging positions of the X-component and the Y-component of laser light R.

(65) [2-1. Configuration]

(66) While in the prism **5** of the optical system **1** of the first embodiment, the first intermediate imaging position P_x and the second intermediate imaging position P_y lie in the vicinity of one reflection surface, the optical system **1E** of this embodiment has the first intermediate imaging position P_x or the second intermediate imaging position P_y lying in the respective vicinities of the two reflection surfaces. The optical system **1** of the first embodiment is the same as the optical system **1E** of this embodiment in configuration except for these different points.

(67) A prism **5E** of this embodiment includes the incident surface **11** as the first surface, the first reflection surface **13** as the second surface, the second reflection surface **17** as the third embodiment, and the exit surface **15** as a fourth surface.

(68) The second reflection surface **17** faces each of the first reflection surface **13** and the exit surface **15**. The second reflection surface **17** reflects the light flux travelling from the first reflection surface **13**, toward the exit surface **15**. The second reflection surface **17** may be of a free curved surface shape with different curvatures in X-direction and Y-direction.

(69) The second intermediate imaging position P_y lies within the second range $Rg2$ from the first reflection surface **13** between the first reflection surface **13** and the second reflection surface **17**. The closer the distance between the second intermediate imaging position P_y and the first reflection surface **13** is, the smaller the size (footprint size) of the light flux at the first reflection surface **13** can be. The smaller the size of the light flux at the first reflection surface **13** is, the less the influence of manufacturing error of the first reflection surface **13** on the light flux can be.

(70) The first intermediate imaging position P_x lies within the second range $Rg2$ from the second reflection surface **17** between the first reflection surface **13** and the second reflection surface **17**. The closer the distance between the first intermediate imaging position P_x and the second reflection surface **17** is, the smaller the size (footprint size) of the light flux at the second reflection surface **17** can be. The smaller the size of the light flux at the second reflection surface **17** is, the less the influence of manufacturing error of the second reflection surface **17** on the light flux can be.

(71) By arranging each of the first intermediate imaging position P_x and the second intermediate imaging position P_y in the vicinity of different reflection surfaces on the optical path in this manner, the first intermediate imaging position P_x and the second intermediate imaging position P_y can be arranged closer to the second reflection surface **17** and the first reflection surface **13**, respectively. This enables further reduction in the influence of manufacturing error of the first reflection surface **13** and the second reflection surface **17**. The influence of dust or scratch of the prism **5E** can be even further reduced.

(72) Reference is then made to FIGS. **13** and **14**. FIG. **13** is a diagram showing a configuration of an optical system **1F** in a variant of the second embodiment. FIG. **14** is a diagram showing respective intermediate imaging positions of the X-component and Y-component of laser light.

(73) The second intermediate imaging position P_y lies, between the first reflection surface **13** and the second reflection surface **17** of a prism **5F**, within the second range $Rg2$ from the first reflection surface **13**. The closer the distance between the second intermediate imaging position P_y and the first reflection surface **13** is, the smaller the size (footprint size) of the light flux at the first reflection surface **13** can be. The smaller the size of the light flux at the first reflection surface **13** is, the less the influence of manufacturing error of the first reflection surface **13** on the light flux can be.

(74) The first intermediate imaging position P_x lies within a third range $Rg3$ from the second reflection surface **17** between the second reflection surface **17** and the exit surface **15**. The third range $Rg3$ may have a length less than one-half, less than one-quarter, or less than one-eighth of an

optical path length L3 from the second reflection surface **17** to the exit surface **15**. The closer the distance between the first intermediate imaging position Px and the second reflection surface **17** is, the smaller the size (footprint size) of the light flux at the second reflection surface **17** can be. The smaller the size of the light flux at the second reflection surface **17** is, the less the influence of manufacturing error of the second reflection surface **17** on the light flux can be. Accordingly, if the third range Rg3 has a length less than one-quarter, instead of less than one-half, of the optical path length L3 from the second reflection surface **17** to the exit surface **15**, the influence of manufacturing error can be further reduced, and if it has a length less than one-eighth instead of less than one-quarter, the influence of manufacturing error can be even further reduced.

(75) An additional reflection surface may be disposed on the optical path between the second reflection surface **17** and the exit surface **15**. In this case, the additionally disposed reflection surface functions as the fourth surface. The prism 5F is the same as the prism 5E in configuration except for the above points.

(76) Reference is then made to FIGS. **15** and **16**. FIG. **15** is a diagram showing a configuration of an optical system **1G** in a variant of the second embodiment. FIG. **16** is a diagram showing respective intermediate imaging positions of the X-component and Y-component of laser light.

(77) The second intermediate imaging position Py lies, between the incident surface **11** and the first reflection surface **13** of a prism 5G, within the first range Rg1 from the first reflection surface **13**. The closer the distance between the second intermediate imaging position Py and the first reflection surface **13** is, the smaller the size (footprint size) of the light flux at the first reflection surface **13** can be. The smaller the size of the light flux at the first reflection surface **13** is, the less the influence of manufacturing error of the first reflection surface **13** on the light flux can be.

(78) The first intermediate imaging position Px lies within the third range Rg3 from the second reflection surface **17** between the second reflection surface **17** and the exit surface **15**. The closer the distance between the first intermediate imaging position Px and the second reflection surface **17** is, the smaller the size (footprint size) of the light flux at the second reflection surface **17** can be. The smaller the size of the light flux at the second reflection surface **17** is, the less the influence of manufacturing error of the second reflection surface **17** on the light flux can be.

(79) An additional reflection surface may be disposed on the optical path between the second reflection surface **17** and the exit surface **15**. In this case, the additionally disposed reflection surface functions as the fourth surface. The prism 5G is the same as the prism 5E in configuration except for the above points. Also in the optical system **1E**, the lens element **7** having different condensing actions in X-direction and Y-direction of the light flux may be arranged between the laser element **3** and the incident surface **11** of the prism 5E.

(80) [2-2. Effects, etc.]

(81) The optical system **1** according to the second embodiment includes the incident surface **11** as the first surface, the first reflection surface **13** as the second surface, the second reflection surface **17** as the third embodiment, and the exit surface **15** as the fourth surface. In the prism 5E, the light flux from the incident surface **11** is reflected by the first reflection surface **13** to head for the second reflection surface **17**, and the light flux from the first reflection surface **13** is reflected by the second reflection surface **17** to head for the exit surface **15**. The prism 5E has: the first intermediate imaging position Px at which the component in X-direction of the light flux incident inside is imaged; and the second intermediate imaging position Py at which the component in Y-direction orthogonal to X-direction of the light flux is imaged. One of the first intermediate imaging position Px and the second intermediate imaging position Py lies within the first range Rg1 from the first reflection surface **13** between the incident surface **11** and the first reflection surface **13**, or lies within the second range Rg2 from the second reflection surface **17** between the first reflection surface **13** and the second reflection surface **17**. The other of the first intermediate imaging position Px and the second intermediate imaging position Py lies within the second range Rg2 from the second reflection surface **17** between the first reflection surface **13** and the second reflection

surface **17**, or lies within the third range Rg3 from the second reflection surface **17** between the second reflection surface **17** and the exit surface **15**. The second range Rg2 has the length less than one-half of the optical path length from the first reflection surface **13** to the second reflection surface **17**. The third range Rg3 has the length less than one-half of the optical path length from the second reflection surface **17** to the exit surface **15**.

(82) Since the two intermediate imaging positions are formed within the prism **5E** based on such a configuration, partial loss of the light flux due to scratch or dust inside the prism **5E** can be reduced. Since at least one of the first intermediate imaging position Px and the second intermediate imaging position Py lies in the vicinity of the second reflection surface **17**, it is possible to diminish the size (footprint size) of the light flux at the second reflection surface **17** to reduce the influence of manufacturing error of the second reflection surface **17** on the light flux.

(83) The two intermediate imaging positions, i.e., the first intermediate imaging position Px and the second intermediate imaging position Py may lie in the respective vicinities of the two reflection surfaces, i.e., the first reflection surface **13** and the second reflection surface **17**, respectively. This enables reduction in size of the light flux reflected by each of the reflection surfaces, making it possible to lessen the influence of manufacturing error of each of the reflection surfaces on the light flux.

Third Embodiment

(84) A third embodiment will then be described with reference to FIGS. **17** and **18**. FIG. **17** is a diagram showing a configuration of an optical system **1H** in the third embodiment. FIG. **18** is a diagram showing respective intermediate imaging positions of the X-component and Y-component of laser light R.

(85) [3-1. Configuration]

(86) As shown in FIG. **17**, the optical system **1H** of this embodiment further includes a third reflection surface **19** in addition to the optical system **1E** of the second embodiment. The optical system **1E** in the second embodiment is the same as the optical system **1H** of this embodiment in configuration except for points described below.

(87) A prism **5H** of this embodiment includes the incident surface **11** as the first surface, the first reflection surface **13** as the second surface, the second reflection surface **17** as the third embodiment, the third reflection surface **19** as the fourth surface, and the exit surface **15** as a fifth surface.

(88) The third reflection surface **19** confronts each of the second reflection surface **17** and the exit surface **15**. The third reflection surface **19** reflects the light flux proceeding from the second reflection surface **17**, toward the exit surface **15**. The third reflection surface **19** may be of a free curved surface shape with different curvatures in X-direction and Y-direction.

(89) The first intermediate imaging position Px lies within the third range Rg3 from the second reflection surface **17** between the second reflection surface **17** and the third reflection surface **19**. The closer the distance between the first intermediate imaging position Px and the second reflection surface **17** is, the smaller the size (footprint size) of the light flux at the second reflection surface **17** can be. The smaller the size of the light flux at the second reflection surface **17** is, the less the influence of manufacturing error of the second reflection surface **17** on the light flux can be.

(90) The second intermediate imaging position Py lies within the second range Rg2 from the first reflection surface **13** between the first reflection surface **13** and the second reflection surface **17**. The closer the distance between the second intermediate imaging position Py and the first reflection surface **13** is, the smaller the size (footprint size) of the light flux at the first reflection surface **13** can be. The smaller the size of the light flux at the first reflection surface **13** is, the less the influence of manufacturing error of the first reflection surface **13** on the light flux can be.

(91) Since the two intermediate imaging positions, i.e., the first intermediate imaging position Px and the second intermediate imaging position Py lie in the respective vicinities of the first reflection surface **13** and the second reflection surface **17**, respectively, it is possible to lessen the

influence of manufacturing error of the first reflection surface **13** and the second reflection surface **17** on the reflected light flux. Accordingly, only the third reflection surface **19** may be intensively managed in manufacture, making it possible to reduce the number of reflection surfaces to be intensively managed, to thereby reduce labor in manufacturing. The second intermediate imaging position **Py** and the first intermediate imaging position **Px** are arranged at positions posterior to reflection at the first reflection surface **13** and the second reflection surface **17**, respectively. By doing so, the second intermediate imaging position **Py** and the first intermediate imaging position **Px** are brought closer the third reflection surface **19** as much as possible, diminishing the size (footprint size) of the light flux at the third reflection surface **19**. Also in the optical system **1H**, the lens element **7** having different condensing actions in X-direction and Y-direction of the light flux may be arranged between the laser element **3** and the incident surface **11** of the prism **5H**.

(92) [3-2. Effects, etc.]

(93) The optical system **1H** according to the third embodiment includes the incident surface **11** as the first surface, the first reflection surface **13** as the second surface, the second reflection surface **17** as the third embodiment, the third reflection surface **19** as the fourth surface, and the exit surface **15** as the fifth surface. In the prism **5H**: the light flux from the incident surface **11** is reflected by the first reflection surface **13** to head for the second reflection surface **17**; the light flux from the first reflection surface **13** is reflected by the second reflection surface **17** to head for the third reflection surface **19**; and the light flux from the second reflection surface **17** is reflected by the third reflection surface **19** to head for the exit surface **15**. The prism **5E** has: the first intermediate imaging position **Px** at which the component in X-direction of the light flux incident inside is imaged; and the second intermediate imaging position **Py** at which the component in Y-direction orthogonal to X-direction of the light flux is imaged. The first intermediate imaging position **Px** lies within the third range **Rg3** from the second reflection surface **17** between the second reflection surface **17** and the third reflection surface **19**. The third range **Rg3** has the length less than one-half of the optical path length from the second reflection surface **17** to the third reflection surface **19**. The second intermediate imaging position **Py** lies within the second range **Rg2** from the first reflection surface **13** between the first reflection surface **13** and the second reflection surface **17**.

(94) Since the two intermediate imaging positions, i.e., the first intermediate imaging position **Px** and the second intermediate imaging position **Py** are formed within the prism **5H** based on such a configuration, partial loss of the light flux due to scratch or dust inside the prism **5H** can be reduced. Since the first intermediate imaging position **Px** is arranged in the vicinity of the second reflection surface **17** while the second intermediate imaging position **Py** is arranged in the vicinity of the first reflection surface **13**, it is possible to diminish the size (footprint size) of the light flux at each of the first reflection surface **13** and the second reflection surface **17** to reduce the influence of manufacturing error of the first reflection surface **13** and the second reflection surface **17** on the light flux.

Other Embodiments

(95) As above, the first to third embodiments have been described as exemplifications of the techniques disclosed in the present application. However, the techniques in the present disclosure are not limited thereto, and are applicable to any embodiments undergoing alterations, permutations, additions, omissions, etc. It is also possible to combine the constituent elements described in the first to third embodiment, into a new embodiment.

(96) Although the exit surfaces **15** of the prisms **5** to **5H** are convex in the first to third embodiments, this is not limitative. The exit surfaces **15** of the prisms **5** to **5H** may not be convex, for example, may be planar.

(97) Although in the variant of the second embodiment shown in FIG. **15**, the first intermediate imaging position **Px** lies within the third range **Rg3** from the second reflection surface **17** between the second reflection surface **17** and the exit surface **15**, this is not limitative. As shown in FIG. **19**,

in an optical system **1K**, the first intermediate imaging position **Px** may lie, between the first reflection surface **13** and the second reflection surface **17** of a prism **5K**, within the second range **Rg2** from the second reflection surface **17**. The closer the distance between the first intermediate imaging position **Px** and the second reflection surface **17** is, the smaller the size (footprint size) of the light flux at the second reflection surface **17** can be. The smaller the size of the light flux at the second reflection surface **17** is, the less the influence of manufacturing error of the second reflection surface **17** on the light flux can be.

(98) As above, the embodiments have been described as exemplifications of the techniques in the present disclosure. To that end, the accompanying drawings and detailed description have been provided. Accordingly, the constituent elements described in the accompanying drawings and detailed description may include not only constituent elements essential for solving the problems but also constituent elements, not essential for problem solving, for exemplifying the above techniques. Hence, those inessential constituent elements should not be recognized as being essential immediately from the fact that those inessential constituent elements are described in the accompanying drawings or detailed description.

(99) Since the above embodiments are intended to exemplify the techniques in the present disclosure, it is possible in claims or their equivalences to make various alterations, permutations, additions, omissions, etc.

SUMMARY OF THE EMBODIMENTS

(100) (1) The optical system of the present disclosure is an optical system including a prism that has the first surface, the second surface, and the third surface, with the light flux from the first surface being reflected by the second surface to head for the third surface. The prism has: the first intermediate imaging position at which the component in the first direction of the light flux incident inside is imaged; and the second intermediate imaging position at which the component in the second direction orthogonal to the first direction of the light flux is imaged, different from the first intermediate imaging position. At least one of the first intermediate imaging position and the second intermediate imaging position lies within the first range from the second surface between the first surface and the second surface, or lies within the second range from the second surface between the second surface and the third surface. The first range has the length less than one-half of the optical path length from the first surface to the second surface, and the second range has the length less than one-half of the optical path length from the second surface to the third surface.

(101) Due to two positions at each of which the light flux is intermediately imaged within the prism, the influence of scratch in the prism can be dispersed. Due to arrangement of either the first intermediate imaging position or the second intermediate imaging position in the vicinity of the second surface, the size of the light flux reflected by the second surface can be diminished to lessen the influence of manufacturing error of the second surface on the light flux.

(102) (2) In the optical system of (1), one of the first intermediate imaging position and the second intermediate imaging position lies within the first range from the second surface between the first surface and the second surface, while the other of the first intermediate imaging position and the second intermediate imaging position lies within the second range from the second surface between the second surface and the third surface.

(103) As a result, one of the first intermediate imaging position and the second intermediate imaging position lies on the optical path toward the first surface from the second surface, while the other of the first intermediate imaging position and the second intermediate imaging position lies on the optical path toward the third surface from the second surface, allowing both the first intermediate imaging position and the second intermediate imaging position to come closer to the vicinity of the second surface. This enables further reduction in size of the light flux reflected by the second surface, to further lessen the influence of manufacturing error of the second surface on the light flux. By arranging the first intermediate imaging position and the second intermediate imaging position within different respective ranges from the second surface, the influence of dust

or scratch in the prism can be further reduced.

(104) (3) In the optical system of (1), both the first intermediate imaging position and the second intermediate imaging position lie within the first range from the second surface between the first surface and the second surface, or lie within the second range from the second surface between the second surface and the third surface.

(105) This allows both the first intermediate imaging position and the second intermediate imaging position to lie in the vicinity of the second surface, enabling the size of the light flux reflected by the second surface to be even smaller, to further reduce the influence of manufacturing error of the second surface on the light flux.

(106) (4) In the optical system of any one of (1) to (3), the first range has the length less than one-quarter of the optical path length from the first surface to the second surface, while the second range has the length less than one-quarter of the optical path length from the second surface to the third surface.

(107) Since the intermediate imaging positions lie even closer to the second surface, the size of the light flux reflected by the second surface can be further diminished, achieving further reduction in influence of manufacturing error of the second surface on the light flux.

(108) (5) In the optical system of any one of (1) to (3), the first range has the length less than one-eighth of the optical path length from the first surface to the second surface, while the second range has the length less than one-eighth of the optical path length from the second surface to the third surface.

(109) Since the intermediate imaging positions lie yet even closer to the second surface, the size of the light flux reflected by the second surface can be even further diminished, achieving still further reduction in influence of manufacturing error of the second surface on the light flux.

(110) (6) In the optical system of any one of (1) to (5), the first surface is the incident surface.

(111) (7) The optical system of (6) includes the lens element having different condensing actions in the first direction and the second direction, with the light flux leaving the lens element being incident on the incident surface. This facilitates design of the intermediate imaging positions within the prism.

(112) (8) In the optical system of any one of (1) to (7), at least one of the first surface, the second surface, and the third surface has a free curved surface shape.

(113) (9) The optical system of the present disclosure is an optical system that includes the prism having the first surface, the second surface, the third surface, and the fourth surface, with the light flux from the first surface being reflected by the second surface to head for the third surface, the light flux from the second surface being reflected by the third surface to head for the fourth surface. The prism has: the first intermediate imaging position at which the component in the first direction of the light flux incident inside is imaged; and the second intermediate imaging position at which the component in the second direction orthogonal to the first direction of the light flux is imaged, different from the first intermediate imaging position. One of the first intermediate imaging position and the second intermediate imaging position lies within the first range from the second surface between the first surface and the second surface, or lies within the second range from the second surface between the second surface and the third surface. The other of the first intermediate imaging position and the second intermediate imaging position lies within the second range from the third surface between the second surface and the third surface, or lies within the third range from the third surface between the third surface and the fourth surface. The first range has the length less than one-half of the optical path length from the first surface to the second surface. The second range has the length less than one-half of the optical path length from the second surface to the third surface. The third range has the length less than one-half of the optical path length from the third surface to the fourth surface.

(114) (10) In the optical system of (9), one of the first intermediate imaging position and the second intermediate imaging position lies within the second range from the second surface between the

second surface and the third surface, while the other of the first intermediate imaging position and the second intermediate imaging position lies within the third range from the third surface between the third surface and the fourth surface. This makes it possible to arrange one of the first intermediate imaging position and the second intermediate imaging position in the vicinity of the second surface and to arrange the other of the first intermediate imaging position and the second intermediate imaging position in the vicinity of the third surface. By arranging the two intermediate imaging positions in the respective vicinities of the different surfaces, the influence of manufacturing error of each of the surfaces can be further reduced. The influence of dust or scratch in the prism can also be further reduced.

(115) (11) In the optical system of (9), one of the first intermediate imaging position and the second intermediate imaging position lies within the first range from the second surface between the first surface and the second surface, while the other of the first intermediate imaging position and the second intermediate imaging position lies within the third range from the third surface between the third surface and the fourth surface. This makes it possible to arrange one of the first intermediate imaging position and the second intermediate imaging position in the vicinity of the second surface and to arrange the other of the first intermediate imaging position and the second intermediate imaging position in the vicinity of the third surface. By arranging the two intermediate imaging positions in the respective vicinities of the different surfaces, the influence of manufacturing error of each of the surfaces can be further reduced. The influence of dust or scratch in the prism can also be further reduced.

(116) (12) In the optical system of (9), one of the first intermediate imaging position and the second intermediate imaging position lies within the first range from the second surface between the first surface and the second surface, while the other of the first intermediate imaging position and the second intermediate imaging position lies within the second range from the third surface between the second surface and third surface. This makes it possible to arrange one of the first intermediate imaging position and the second intermediate imaging position in the vicinity of the second surface and to arrange the other of the first intermediate imaging position and the second intermediate imaging position in the vicinity of the third surface. By arranging the two intermediate imaging positions in the respective vicinities of the different surfaces, the influence of manufacturing error of each of the surfaces can be further reduced. The influence of dust or scratch in the prism can also be further reduced.

(117) (13) In the optical system of (9), one of the first intermediate imaging position and the second intermediate imaging position lies within the second range from the second surface between the second surface and the third surface, while the other of the first intermediate imaging position and the second intermediate imaging position lies within the second range from the third surface between the second surface and third surface. This makes it possible to arrange one of the first intermediate imaging position and the second intermediate imaging position in the vicinity of the second surface and to arrange the other of the first intermediate imaging position and the second intermediate imaging position in the vicinity of the third surface. By arranging the two intermediate imaging positions in the respective vicinities of the different surfaces, the influence of manufacturing error of each of the surfaces can be further reduced. The influence of dust or scratch in the prism can also be further reduced.

(118) (14) In the optical system of (9): the first range has the length less than one-quarter of the optical path length from the first surface to the second surface; the second range has the length less than one-quarter of the optical path length from the second surface to the third surface; and the third range has the length less than one-quarter of the optical path length from the third surface to the fourth surface.

(119) (15) In the optical system of (9): the first range has the length less than one-eighth of the optical path length from the first surface to the second surface; the second range has the length less than one-eighth of the optical path length from the second surface to the third surface; and the third

range has the length less than one-eighth of the optical path length from the third surface to the fourth surface.

(120) (16) In the optical system of any one of (9) to (15), the first surface is the incident surface.

(121) (17) The optical system of (16) includes the lens element having different condensing actions in the first direction and the second direction, with the light flux leaving the lens element being incident on the incident surface. This facilitates design of the intermediate imaging positions within the prism.

(122) (18) In the optical system of any one of (9) to (17), at least one of the first surface, the second surface, the third surface, and the fourth surface has a free curved surface shape.

(123) (19) In the optical system of any one of (1) to (18), the refractive index n of the prism satisfies the relationship of $n < 1.8$. This enables the influence of manufacturing error of the refraction surface of the prism to be reduced.

(124) (20) In the optical system of any one of (1) to (19), the Abbe number v_d of the prism satisfies the relationship of $v_d > 40$. This achieves reduction in chromatic aberration of the prism.

(125) (21) In the optical system of any one of (1) to (20), let M_x be the horizontal magnification in the first direction and let M_y be the horizontal magnification in the second direction, the relationship of $M_y/M_x > 1.1$ is satisfied. This results in an increased optical path length difference between the first intermediate imaging position and the second intermediate imaging position, allowing alleviation of the influence of scratch or dust inside the prism.

(126) (22) In the optical system of any one of (1) to (21), let ΔP be the optical path length difference between the first intermediate imaging position and the second intermediate imaging position and let L be the optical path length from the first surface to the second surface, the relationship of $\Delta P/L > 0.05$ is satisfied. It is consequently possible to properly set the optical path length difference between the first intermediate imaging position and the second intermediate imaging position to alleviate the influence of scratch or dust inside the prism.

(127) (23) The optical system of any one of (1) to (22) further includes the laser element irradiating laser light onto the incident surface of the prism, with the first direction being the major axis direction of the pupil diameter of laser light emitted from the laser element, the second direction being the minor axis direction of the pupil diameter of laser light emitted from the laser element, the first direction and the second direction being orthogonal to each other, the second intermediate imaging position lying closer to the incident surface than the first intermediate imaging position does.

(128) The present disclosure is applicable to an optical system that uses a dioptric system such as a prism.

EXPLANATIONS OF LETTERS OR NUMERALS

(129) **1, 1A, 1B, 1C, 1D, 1E, 1F, 1G, 1H, 1K** optical system **3** laser element **3a, 3b** pupil diameter **5, 5A, 5B, 5C, 5D, 5E, 5F, 5G, 5H, 5K** prism **11** incident surface **13** first reflection surface **15** exit surface **17** second reflection surface **19** third reflection surface **L1, L2, L3** optical path length **Px** first intermediate imaging position **Py** second intermediate imaging position **R** laser light

Claims

1. An optical system comprising a prism that includes a first surface, a second surface, and a third surface, with a light flux from the first surface being reflected by the second surface to head for the third surface, the prism having: a first intermediate imaging position at which a component in a first direction of the light flux incident inside is imaged; and a second intermediate imaging position at which a component in a second direction orthogonal to the first direction of the light flux is imaged, different from the first intermediate imaging position, at least one of the first intermediate imaging position and the second intermediate imaging position lying within a first range from the second surface between the first surface and the second surface or lying within a

second range from the second surface between the second surface and the third surface, the first range having a length less than one-half of an optical path length from the first surface to the second surface, the second range having a length less than one-half of the optical path length from the second surface to the third surface.

2. The optical system of claim 1, wherein one of the first intermediate imaging position and the second intermediate imaging position lies within the first range from the second surface between the first surface and the second surface, while the other of the first intermediate imaging position and the second intermediate imaging position lies within the second range from the second surface between the second surface and the third surface.

3. The optical system of claim 1, wherein both of the first intermediate imaging position and the second intermediate imaging position lie within the first range from the second surface between the first surface and the second surface or lie within the second range from the second surface between the second surface and the third surface.

4. The optical system of claim 1, wherein the first range has a length less than one-quarter of the optical path length from the first surface to the second surface, and wherein the second range has a length less than one-quarter of the optical path length from the second surface to the third surface.

5. The optical system claim 1, wherein the first range has a length less than one-eighth of the optical path length from the first surface to the second surface, and wherein the second range has a length less than one-eighth of the optical path length from the second surface to the third surface.

6. The optical system claim 1, wherein the first surface is an incident surface.

7. The optical system of claim 6, comprising a lens element that has different condensing actions in the first direction and the second direction, wherein the light flux leaving the lens element is incident on the incident surface.

8. The optical system of claim 1, at least one of the first surface, the second surface, and the third surface has a free curved surface shape.

9. The optical system of claim 1, wherein the prism has a refractive index n satisfying a relationship of $n < 1.8$.

10. The optical system of claim 1, wherein the prism has an Abbe number v_d satisfying a relationship of $v_d > 40$.

11. The optical system of claim 1, let M_x be a horizontal magnification in the first direction and let M_y be a horizontal magnification in the second direction, a relationship of $M_y/M_x > 1.1$ is satisfied.

12. The optical system of claim 1, let ΔP be an optical path length difference between an optical path length from the first surface to the first intermediate imaging position and an optical path length from the first surface to the second intermediate imaging position and let L be the optical path length from the first surface to the second surface, a relationship of $\Delta P/L > 0.05$ is satisfied.

13. The optical system of claim 1, further comprising a laser element irradiating laser light onto an incident surface of the prism, wherein the first direction is a major axis direction of a pupil diameter of laser light emitted from the laser element, while the second direction is a minor axis direction of the pupil diameter of the laser light emitted from the laser element, the first direction and the second direction being orthogonal to each other, and wherein the second intermediate imaging position lies closer to the incident surface than the first intermediate imaging position does.

14. An optical system comprising a prism that includes a first surface, a second surface, a third surface, and a fourth surface, with a light flux from the first surface being reflected by the second surface to head for the third surface, the light flux from the second surface being reflected by the third surface to head for the fourth surface, the prism having: a first intermediate imaging position at which a component in a first direction of the light flux incident inside is imaged; and a second intermediate imaging position at which a component in a second direction orthogonal to the first direction of the light flux is imaged, different from the first intermediate imaging position, one of the first intermediate imaging position and the second intermediate imaging position lying within a

first range from the second surface between the first surface and the second surface or lying within a second range from the second surface between the second surface and the third surface, the other of the first intermediate imaging position and the second intermediate imaging position lying within the second range from the third surface between the second surface and the third surface or lying within a third range from the third surface between the third surface and the fourth surface, the first range having a length less than one-half of an optical path length from the first surface to the second surface, the second range having a length less than one-half of the optical path length from the second surface to the third surface, the third range having a length less than one-half of the optical path length from the third surface to the fourth surface.

15. The optical system of claim 14, wherein one of the first intermediate imaging position and the second intermediate imaging position lies within the second range from the second surface between the second surface and the third surface, and wherein the other of the first intermediate imaging position and the second intermediate imaging position lies within the third range from the third surface between the third surface and the fourth surface.

16. The optical system of claim 14, wherein one of the first intermediate imaging position and the second intermediate imaging position lies within the first range from the second surface between the first surface and the second surface, and wherein the other of the first intermediate imaging position and the second intermediate imaging position lies within the third range from the third surface between the third surface and the fourth surface.

17. The optical system of claim 14, wherein one of the first intermediate imaging position and the second intermediate imaging position lies within the first range from the second surface between the first surface and the second surface, and wherein the other of the first intermediate imaging position and the second intermediate imaging position lies within the second range from the third surface between the second surface and third surface.

18. The optical system of claim 14, wherein one of the first intermediate imaging position and the second intermediate imaging position lies within the second range from the second surface between the second surface and the third surface, and wherein the other of the first intermediate imaging position and the second intermediate imaging position lies within the second range from the third surface between the second surface and third surface.
